

CLaP - Contrast, Label, Predict: a quest for cheaper labeling in 3D human pose estimation

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Abstract

Human pose estimation (HPE) is a pivotal task in computer vision with applications spanning a wide range of domains, such as sports analytics, rehabilitation, performance capture and many more. However, obtaining labeled datasets for 3D pose estimation remains costly and resource intensive. To address this challenge, we propose a novel pipeline that uses contrastive learning to reduce labeling requirements while maintaining adequate performance. Our method employs unsupervised fine-tuning of pre-trained ResNet backbones on unannotated multiview data acquired in a skiing scenario. The learned representations are then utilized to strategically select a minimal, yet diverse subset of data for labeling, which is subsequently used for supervised training. We demonstrate the effectiveness of this approach using three contrastive paradigms, namely SimCLR, MoCo, and SimSiam, evaluating their impact on data efficiency and model performance on the SkiPose dataset. Our results indicate that contrastive learning can significantly reduce labeling costs while retaining good pose estimation results, making it a promising solution for resource-constrained applications. Code is available at [mmlab-cv.github.io/CLaP](https://github.com/mmlab-cv/CLaP).

1. Introduction

Human pose estimation (HPE) is a critical task in computer vision, with applications in areas like sports analysis, where understanding human movement is essential. This research focuses on the challenge of 3D human pose estimation in resource-constrained settings, where labeled data is scarce and costly to obtain.

The core motivation here is straightforward: labeling data is expensive, both in time and resources. For many organizations, the cost of annotating large datasets is pro-

hibitive, leading to a natural question — *how can we achieve high-quality results without the need for exhaustive labeling?* Instead of labeling everything, the goal is to use what is already available more effectively. This is especially relevant in the context of 3D human pose estimation, where the cost of obtaining labeled data is particularly high due to the need for precise 3D annotations, which are often obtained through expensive motion capture systems or manual labeling, both of which are time-consuming and labor-intensive.

This is where contrastive learning comes in. By learning representations directly from unlabeled data, these methods reduce the dependence on large labeled datasets. The idea is to extract meaningful features from the data itself, enabling better performance even in scenarios with limited annotations. Through this approach, our objective is to demonstrate the effectiveness of contrastive learning in enhancing pose estimation capabilities in lower-resource settings, paving the way for more cost-effective solutions in the field.

We propose a novel pipeline that leverages contrastive learning to reduce the need for extensive labeling in 3D human pose estimation. Our method involves unsupervised fine-tuning of pre-trained ResNet backbones on unannotated multiview data, followed by strategic selection of a minimal, yet diverse subset of data for labeling. This subset is then used for supervised training, enabling the model to generalize effectively across different poses and views. We evaluate the performance of this approach using three contrastive paradigms—SimCLR, MoCo, and SimSiam—on the SkiPose dataset, showcasing the benefits of contrastive learning in reducing labeling costs while maintaining high-quality pose estimation results.

In the next section, we will explore the related works in literature. Taking inspiration from these, we will first define a baseline, as well as a more refined training paradigm that allows us to show the effectiveness of using contrastive

learning for labeling purposes. Finally, we will showcase our models' improved results over our baseline method, both qualitatively and quantitatively.

2. Related Work

In the field of Human Pose Estimation, contrastive learning is showing great promise as a way to harness the vast amounts of unlabeled data available. The idea is simple but powerful: you train the system to recognize that similar inputs should have similar representations, while different inputs should be distinct. By maximizing agreement between positive pairs and minimizing it for negative pairs, we can teach the model to extract meaningful, robust pose features. It is an elegant approach to getting more out of both labeled and unlabeled data, and it has the potential to push the boundaries of what we can do with data-efficient learning.

2.1. Contrastive Learning

One of the first successful approaches in contrastive learning optimizes the similarity between different augmented views of the same image while contrasting them against a large pool of negative samples. This is achieved using the InfoNCE [12] loss function, though this method relies on large batch sizes and significant computational resources. These requirements can limit its practical applicability, especially in constrained computational environments where unlabeled data may be more readily available. This method was first implemented in **SimCLR** [1], despite having some limitations, such as its resource-intensive hardware requirements during training given the need for a high batch size.

Differently to SimCLR, **MoCo** [4] introduced a momentum encoder that leverages a dynamic queue of negative samples, enabling the use of a more diverse set of negative instances during training without the need for fully labeled data. This method is particularly suited for scenarios like HPE, where partial labels, such as those obtainable from synchronized camera feeds, can support robust representation learning. MoCo's momentum-encoder-based approach enhances the model's ability to learn discriminative features while using fewer labeled samples.

To address the challenge of collapsing representations, the authors of **SimSiam** [2] proposed an asymmetric architecture, adding a projection MLP head, or predictor, on one branch of the network to align positive pairs of representations. This approach is effective in learning representations even with limited labeled data, making it attractive for settings where labels are scarce. However, its focus on aligning single-modality representations limits its effectiveness in multi-modal tasks, such as HPE, where integrating data from sources like RGB images and pose coordinates

is essential. SimSiam's architecture has thus been applied mainly in single-modality alignment tasks.

2.2. Human Pose Estimation

In the field of HPE, several attempts have been made to use contrastive learning, both leveraging labeled and unlabeled data. Following the success of language models such as CLIP, **UniHPE** [21] constructs the pose representations by leveraging the ability to align multiple modalities in a shared embedding space. This approach employs a two-stage contrastive alignment method: first, it aligns 2D and 3D pose representations, and then it integrates RGB images. Using a singular value-based contrastive loss, it enables optimized multi-modal feature alignment in a shared feature space, significantly enhancing HPE performance. This technique is particularly useful in scenarios requiring the fusion of multiple data modalities, though it still relies on labeled images for effective alignment, which can limit its applicability in low-label scenarios.

On the other hand, **HandCLR** [9] leverages contrastive learning to pre-train on unlabeled data, enabling it to utilize large-scale datasets of in-the-wild hand images without annotations. This model focuses on learning valuable hand pose representations by distinguishing between similar and dissimilar hand poses in diverse conditions.

Beyond human pose estimation, contrastive learning has been employed to address related challenges, such as resolution mismatch in HPE networks. Some works for example, leveraged contrastive learning to address resolution challenges, with the former ensuring scale consistency in resolution-aware models and the latter integrating local similarity and global features to enhance robustness in low-resolution scenarios [19] [6].

In a different approach, **GenPoCCL** [7] introduces a multi-positive contrastive loss designed to align features across pose-consistent image pairs. This method leverages controllable image generation to produce images with identical poses but varying appearances, treating these pairs as positive examples.

2.3. Active learning

Active learning has been explored to minimize the cost and effort of annotating data while maximizing model performance, especially in complex tasks where labeling is expensive and time-consuming like for Human Pose Estimation (HPE).

A big problem for machine learning as a whole is out of distribution samples. Many works try to address this problem using different techniques. **VL4Pose** [15], for example, employs a Bayesian Network to model skeletal structure constraints and identify out-of-distribution samples for annotation.

Consistency-Weighted Core-Set Approach [20] introduces a generalized core-set framework that integrates uncertainty and representativeness in data selection. This method ensures a balanced trade-off between diverse data representation and high-impact annotation.

Multiple Peak Entropy (MPE) [10] focuses on uncertainty sampling, identifying ambiguous images where pose estimations exhibit multiple weak local peaks in predicted heatmaps. Other approaches try to use a more convoluted approach, trying to learn the best policies to select the new samples to label. **Meta-Agent Teaming Active Learning** (MATAL) [3] reframes active learning as a Markov Decision Process (MDP), leveraging reinforcement learning to discover optimal sampling strategies. MATAL introduces a multi-agent teamwork framework where each agent collaboratively selects informative images, ensuring diversity and individual informativeness in the selected batch.

Surprisingly, little work combines contrastive learning with guiding sample selection in data-scarce domains. Our approach shows how it can not only learn features but also prioritize labeling efforts efficiently.

3. Methodology

Our goal is to design a pipeline capable of leveraging semi-supervised data, particularly datasets containing multiple aligned views of the same pose. These datasets, such as Panoptic [8] and SkiPose [16] [13] [14], are already available and provide an excellent foundation for our approach.

The idea is to use this easily obtainable data to learn robust representations through contrastive learning. By doing so, we can better understand the underlying data distribution, enabling a more informed and selective approach to annotation. Rather than annotating indiscriminately, this process allows us to focus on the most relevant samples, ensuring that the expensive labeling process is used as efficiently as possible to train a high-performing pose estimation model.

The training pipeline we devised first employs an unsupervised contrastive step on data without looking at labels to develop meaningful representations for the problem and domain at hand by leveraging easily acquired data. Then it uses the learned representations to select a subset of the images to be labeled which are later used to train a pose estimation head, which will have by construction a balanced and meaningful coverage of the whole data distribution we are interested in working in. The whole process is illustrated in Figure 1.

3.1. Contrastive Finetuning

To fine-tune our image encoder, we begin with a pre-trained ResNet model on the ImageNet dataset. The method we design is a slight revision of the contrastive paradigm, where, instead of using augmentations of the same image to

align the encoder, we use the images from multiple views of the same pose to be aligned in the model’s representation space. As already mentioned, we decided to use the SkiPose dataset which allows us to use views from six different cameras.

In the standard contrastive learning paradigm, a single augmented training sample is contrasted with all the others in the batch. However, in the context of human pose estimation, this might not be optimal as we would be probably contrasting multiple poses that are the same but captured at different moments in time or from a different viewpoint. In some scenarios, this might not affect the results, as the networks receive a diverse sample of poses and the loss is still able to drive the model to a meaningful representation. However, this assumption breaks down for datasets centered on a specific domain, such as skiing, where the pose diversity is inherently limited to actions relevant to the sport. This is especially true for a sport like skiing.

To address this challenge, we tested different model paradigms to explore various approaches and identify the best balance for learning robust representations:

- **SimCLR**: Uses both positive and negative pairs to build its representations
- **MoCo**: Uses a momentum encoder for the negative samples, allowing to contrast data outside the current batch.
- **SimSiam**: We only use positive pairs to avoid contrasting the same pose, leveraging the ImageNet pre-training and asymmetric architecture to avoid collapse.

3.2. Training samples selection

The resulting models are expected to learn meaningful representations that can consistently score the same pose viewed from multiple angles. This should not only improve performance in the pose estimation task but also produce a well-structured encoding space where similar poses are naturally clustered closer together. Such behavior is crucial for leveraging the encoder’s knowledge to enhance both the performance and efficiency of using labeled data. We want to leverage the encoder’s acquired knowledge by selectively sampling only a portion of the dataset samples, to maximize the efficiency of our pre-training scheme.

The selection starts with clustering using the KNN clustering algorithm. Each sample is assigned to a cluster based on the distance to a cluster centroid. The output from this step should cluster similar poses together, regardless of their viewing angle. We exploit this separation to sample the training examples equally from the sample space, to be used in the next step. This sampling extracts a subset of the original data that evenly and meaningfully covers the encoding space, allowing the downstream model to generalize well across different poses and views.

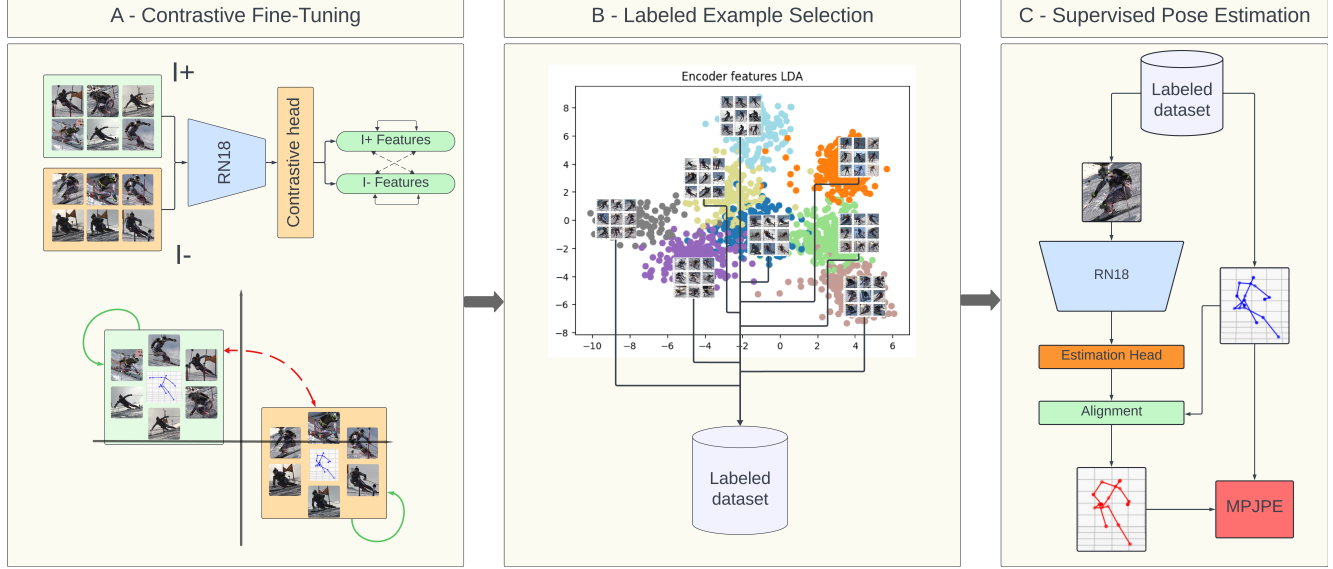


Figure 1. Overview of our three-step process: (A) unsupervised fine-tuning of a Resnet18 backbone using contrastive learning, (B) clustering-based dataset sampling at training-time, and (C) Human Pose Estimation on the frozen contrastive encoder.

3.3. Pose estimation

The resulting subset of training examples is divided into training and evaluation splits to be used in the supervised learning step for human pose estimation.

To evaluate the quality of the representations learned through contrastive fine-tuning, we employ a simple pose estimation head. This simplicity is intentional, as it allows us to focus on the impact of the learned representations on the pose estimation task. More complex decoding heads may lead to better results, but that is not the focus of this work, but rather meaningful representations. To further isolate the effect of the encoder’s learned features, we freeze the weights of the encoder and backpropagate the error only through the estimation head. This approach ensures that any improvements in pose estimation performance can be attributed to the quality of the learned representations.

This model is only required to learn the joint positions agnostic from the view the encoder receives, which should have collapsed during the contrastive fine-tuning. Thus, the loss is computed by first aligning the GT pose and the estimation by centering both and rotating the latter to minimize the mean square error, and later computing the MPJPE:

$$L_{estim} = \frac{1}{N_B} \frac{1}{N_J} \sum_{b,j} \|p_{b,j} - \hat{p}_{b,j}\| \quad (1)$$

Where N_B is the number of samples in a batch and N_J is the number of joints in the pose.

4. Experiments

We evaluate the impact of using pre-trained and fine-tuned models on our training data. We choose the ResNet18 model [5] as our primary image encoder and test it on the SkiPose dataset [16] [13] [14], which includes images from six cameras distributed along a ski course, annotated with both 2D and 3D poses.

We conduct contrastive fine-tuning with the following settings:

- SimCLR: batch size of 128, learning rate for the encoder at 0.1 and for the projection head at 0.5, temperature set to 0.5, with a cosine annealing learning rate scheduler ($T_{max}=30$, $\eta_{min}=1e-6$), training with the LARS optimizer for 30 epochs.
- MoCo: batch size of 128, a key queue size of 512, same learning rates and scheduler settings used for SimCLR, training with the SGD optimizer for 30 epochs.
- SimSiam: same batch size, learning rates, temperature, and scheduler settings as SimCLR, training with the SGD optimizer for 30 epochs.

For sample selection, we clustered the data into 20 groups and progressively reduced the samples from 50% to 10%. We also compared this method to a baseline that randomly samples from the training data.

As for the pose estimation head, we designed two-layer MLP with a GeLU activation function and a hidden layer of size 128, normalized using Gaussian noise. An illustration

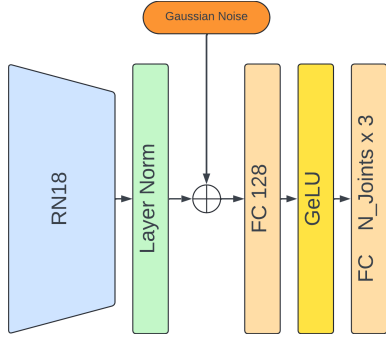


Figure 2. Our simple Human Pose Estimation head architecture: a two-layer MLP with GeLU activation and Gaussian noise normalization.

of the architecture during training is shown in Fig. 2. This model was trained on 80% of the selected training subset, with the remaining 20% used for validation. The training uses an AdamW optimizer with $1e-3$ LR, patience 1 and multiplier 0.1 for LR scheduling, and patience 2 on the validation set as early stopping.

5. Results

We evaluate the effectiveness of our approach across a range of experimental settings, comparing it against a baseline model.

We start by discussing the outcomes of the contrastive fine-tuning, using metrics that should give us some insight into the learned representations of the models. Subsequently, we assess the effectiveness of our method in the pose estimation task by comparing the results obtained by our models with a random sampling baseline. We then conclude with a comprehensive discussion of the overall findings.

5.1. Contrastive Pre-Training

Assessing the quality of the learned embeddings is crucial to understand the effectiveness of the contrastive pre-training methods for 3D human pose estimation. We evaluated the representations learned by different models—ResNet18 (RN18), SimCLR, MoCo, and SimSiam—using several metrics: Silhouette Score with the pose similarity computed using the distance matrix, Spearman and Pearson correlation coefficients between physical pose differences and embedding cosine distances, and K-Nearest Neighbor (K-NN) regression MPJPE. These metrics are computed by first aligning all the poses with the spine upright since we want to eliminate information about rotation and viewing angle, as we trained these models to be view agnostic. Table 1 presents the evaluation metrics for each model.

The Silhouette Score measures the cohesion and separation

of the clusters formed in the encoding space, with higher values indicating better-defined clusters. All models exhibit negative Silhouette Scores, suggesting that the clusters in the high-dimensional embedding space are overlapping. However, SimSiam achieves the highest (least negative) Silhouette Score of -0.113 , indicating slightly better cluster separation compared to the other models. However, we argue that a good silhouette score is not crucial to our downstream task as we ultimately want to have a regression model that might not benefit from a sparser encoding space.

The Spearman and Pearson correlation coefficients evaluate the monotonic and linear relationships, respectively, between the physical differences in poses and the cosine distances between their embeddings. Higher correlation coefficients imply that the embeddings preserve the pose similarities and differences more effectively. SimCLR outperforms the other models in both Spearman (0.279) and Pearson (0.254) correlations, indicating that it captures pose-relevant features in a way that reflects the actual pose distances more accurately.

The K-NN regression MPJPE assesses how well a simple K-NN regressor can predict joint positions based on the encoding. Lower errors indicate more informative embeddings for the pose estimation task. SimSiam achieves the lowest K-NN regression MPJPE of 104 mm, suggesting that its embeddings are most useful for downstream regression tasks.

Comparing these results to the baseline RN18 model, we observe that all contrastive learning methods provide some degree of improvement. SimCLR significantly enhances the correlation between embedding distances and pose differences, which is valuable for tasks that rely on measuring pose similarities, like retrieving similar poses from a database. SimSiam, while not achieving the highest correlations, provides embeddings that are more effective for regression, as indicated by the lowest K-NN regression error.

MoCo’s performance is mixed; it shows moderate improvements over the baseline in the Silhouette Score but does not outperform RN18 in Spearman and Pearson correlations or K-NN regression error. However, better results may be achieved by a more in-depth hyperparameters tuning.

These results demonstrate that different contrastive learning methods offer various advantages. SimCLR is most effective at capturing pose similarities in the encoding space, while SimSiam seems to produce embeddings that are better suited for direct pose regression tasks.

5.2. Pose Estimation

We also compare the performance of the fine-tuned models to a standard ResNet18 model pre-trained on ImageNet on the task of human pose estimation. The evaluation metric

Model	Silhouette Scores $\uparrow$	Spearman Corr $\uparrow$	Pearson Corr $\uparrow$	K-NN reg MPJPE (mm) $\downarrow$
RN18	-0.162	0.154	0.169	106
SimCLR	-0.151	0.279	0.254	119
MoCo	-0.128	0.102	0.136	106
SimSiam	-0.113	0.169	0.199	104

Table 1. Evaluation metrics for different contrastive pre-training methods. Among all the models, SimSiam achieves the best K-NN regression MPJPE and Silhouette Score, while SimCLR shows the highest Spearman and Pearson correlations.

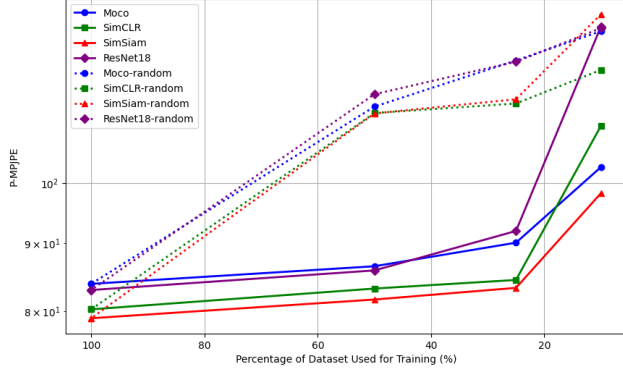


Figure 3. P-MPJPE results for different models and dataset splits. Our model consistently outperforms the random baseline, with SimSiam showing the best performance.

used is MPJPE (Mean Per Joint Position Error), which measures the average Euclidean distance between the predicted and ground-truth joint positions. Lower MPJPE values indicate more accurate 3D human pose estimation. Before calculating the error, we apply a transformation (rotation and translation) to align the predicted pose with the ground truth. The models were trained with varying percentages of the dataset, ranging from 10% to 100%. We also compare the models trained with clustering selection to those trained with random selection. Lastly, we examine the absolute P-MPJPE values for each model and analyze their rescaled performance, evaluating each model’s relative performance compared to its best result achieved using 100% of the training data.

Comparing the model trained with the sampled dataset from the previous step and the same model trained with a random sampling of the dataset, we show that when using our selection strategy the models consistently outperform the random baseline, obtaining a lower P-MPJPE. These results can be seen in Table 2 and in Fig. 3.

At lower data percentages (10% and 25%), ResNet18 shows a higher loss than SimCLR and SimSiam contrastive fine-tuned models. This result confirms the evaluation in Section 5.1, showing that a better encoding space for pose estimation was successfully achieved using contrastive fine-tuning with a semi-supervised paradigm.

As the dataset grows, the gap between RN18 default

Model	Data Split	P-MPJPE (mm)	Ret_P %
RN18	100%	83	100
	50%	86	96.64
	25%	92	90.23
	10%	131	63.19
	Rand 50%	117	71.10
	Rand 25%	124	67.18
	Rand 10%	131	63.36
SimCLR	100%	80	100
	50%	83	96.44
	25%	84	95.01
	10%	110	72.63
	Rand 50%	113	71.03
	Rand 25%	115	69.86
	Rand 10%	122	65.89
MoCo	100%	84	100
	50%	86	96.98
	25%	90	93.08
	10%	103	81.59
	Rand 50%	114	73.44
	Rand 25%	124	67.76
	Rand 10%	130	64.39
SimSiam	100%	79	100
	50%	82	96.80
	25%	83	94.85
	10%	98	80.42
	Rand 50%	113	69.99
	Rand 25%	116	68.28
	Rand 10%	134	58.88

Table 2. P-MPJPE results for different models and dataset splits. The best performance at different sampling percentages is highlighted in bold.

weights and the fine-tuned models shrinks. Still, the superiority of a good selection strategy for the labeled samples becomes evident as the selected dataset significantly outperforms the random baseline.

Figure 4 shows the retained performance (Ret_P) of the models, measured as the relative change in MPJPE with respect to the performance at 100% of the dataset computed

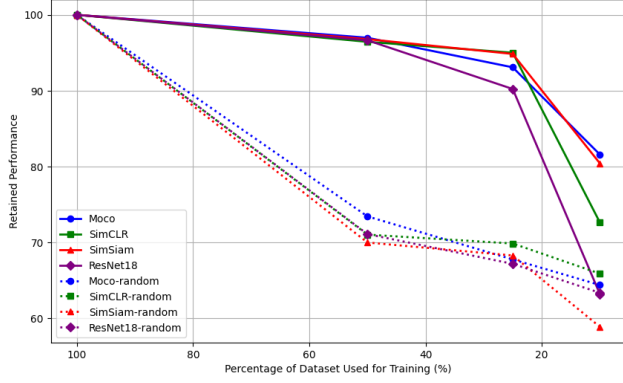


Figure 4. Retained performance of the models at different dataset splits. The contrastive fine-tuned models show a superior retained performance compared to the baseline model.

using the formula below 2. This way, the performance for each model has been rescaled so that the performance at 100% dataset usage is normalized to 100%. This allows us to compare the models’ ability to maintain accuracy as the training data size decreases.

$$\text{Ret_P}_{\text{curr}} = \frac{\text{MPJPE}_{100}}{\text{MPJPE}_{\text{curr}}} 100 \quad (2)$$

From the plot in figure 4, it is evident that the models trained with clustering selection consistently retain a higher level of performance when compared to their random-sampling counterparts. In particular, the contrastive models show a superior retained performance, with the loss curve staying closer to the baseline (100% training data) even at lower dataset percentages. For example, SimSiam and MoCo trained with the clustering selection have a retained performance of more than 80% with only 10% of the dataset. In contrast, the ResNet18 model shows a steeper decline in retained performance as the dataset size decreases, highlighting the importance of the contrastive fine-tuning step for improved generalization.

As shown in Table 3, not many studies have addressed the pose estimation task using the contained Skipose dataset, perhaps due to its limited size and the need for multiple views. The main challenge is to achieve good performance while training on a single view at a time, as this is more common in real-world applications. While MetaPose [17] achieves superior performance, it requires multiple views as input. Notably, our method outperforms CanonPose [18], obtaining competitive results with a fraction of the labeled data.

5.3. Conclusions

We presented “*Contrast, label, predict*”, a 3D human pose estimation approach tailored for resource-constrained environments. It leverages contrastive learning and efficient

Technique	P-MPJPE	Single View
CanonPose [18]	90	Yes
MetaPose [17]	42	No
Ours (full)	79	Yes
Ours (50%)	82	Yes
Ours (25%)	83	Yes

Table 3. Comparison of our method with other works on the SkiPose dataset. Our method achieves competitive results with a fraction of the labeled data, while being trained on a single view at a time.

clustering-based sampling to drastically reduce the need for 3D annotated data at training time.

Our experiments show that contrastive pre-training with SimCLR and SimSiam effectively capture meaningful pose features, consistently outperforming a ResNet18 baseline in the downstream task of pose estimation, highlighting the advantages of semi-supervised methods.

Sampling from the dataset through our clustering-based approach proved to be a valuable strategy, enabling us to focus resources on the most significant samples. This approach consistently led to significantly better results compared to random sampling, while preserving a substantial percentage of the original performance.

Both SimCLR and SimSiam leverage the semi-supervised paradigm effectively, utilizing multiple views to learn meaningful representations despite the relatively limited size of the SkiPose dataset. MoCo, on the other hand, did not outperform ResNet18 when using the full dataset or the 50% selected subset, likely due to suboptimal hyperparameters. However, in scenarios with significantly less labeled data, such as 25% or 10%, MoCo showed improvements over the baseline model, indicating that it did capture beneficial representations during fine-tuning, performing better than SimCLR at 10% but slightly underperforming compared to SimSiam.

Among the methods evaluated, SimSiam is the most data-efficient approach, achieving one of the lowest loss values even when trained on just 25% of the dataset.

6. Future Work

One promising direction is the incorporation of comprehensive semi-supervised training with larger datasets. Extending the quantity of data could significantly enhance feature representation and improve generalization across various scenarios. Access to more comprehensive datasets could be particularly beneficial in addressing edge cases and increasing robustness in real-world applications without requiring big investments, as these datasets can be acquired cheaply.

Another improvement involves scaling the pipeline with

ResNet 18 full - MPJPE 94



SimSiam 18 full - MPJPE 60



SimSiam 18 50% - MPJPE 57



Ground Truth - MPJPE 0

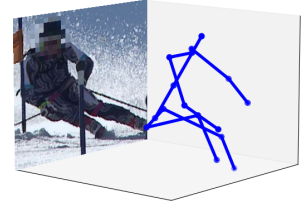
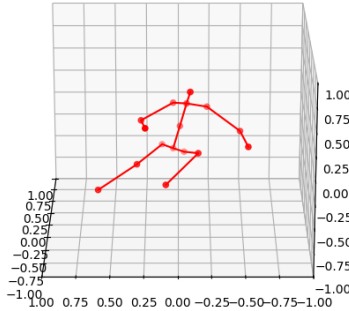
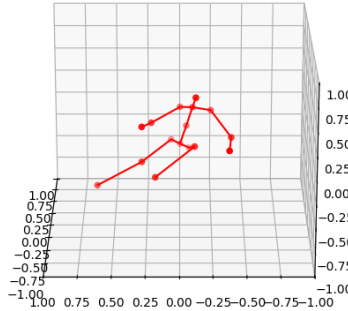


Figure 5. Qualitative results of the pose estimation task on the SkiPose dataset. The images show the ground truth pose (blue) and the predicted pose (red) for different models.

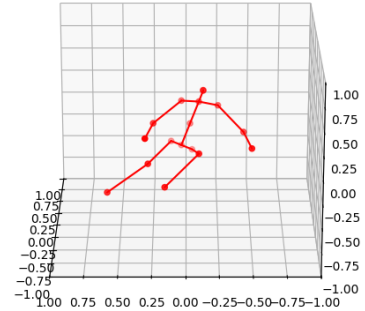
SimSiam 18 full - MPJPE 76



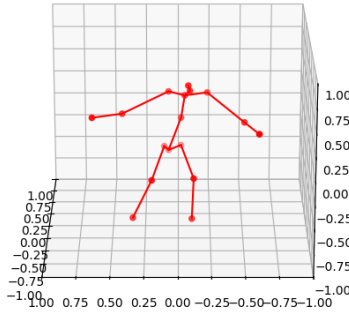
SimSiam 18 25% - MPJPE 84



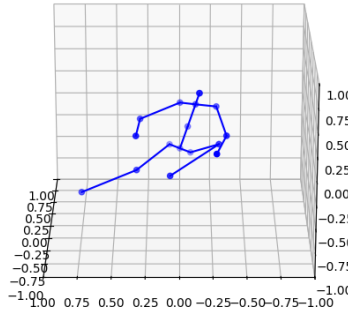
ResNet 18 full - MPJPE 85



SimSiam 18 rand 25% - MPJPE 180



Ground Truth - MPJPE 0



Original Image



Figure 6. Comparison of the pose estimation results for different models on the SkiPose dataset. The images show the ground truth pose (blue) and the predicted pose (red) for different models. As seen from the plots, a random dataset sampling strategy leads to a higher MPJPE and a worse pose compared to the clustering-based selection.

larger backbone architectures and designing more refined estimation heads. Larger models typically offer increased representational capacity, which could translate into better accuracy and robustness in pose estimation tasks. Exploring advanced head designs, such as heatmap regression for joint position refinement allow to capture in a better way the in-

herent uncertainties in pose estimation, as demonstrated in prior work on hand pose estimation [11]. This approach, which moves beyond simplistic representations, has the potential to significantly improve the precision and reliability of joint localization while maintaining computational efficiency.

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